

SALMIYA, Kuwait

WMO: 405850

Lat: 29.350N

Lon: 48.100E

Elev: 26

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.2	51.0	22.6	17.0	56.9	27.1	21.1	59.6	43.1	61.6	34.6	56.5	14.1	310	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	109.5	71.6	107.3	72.4	105.3	72.4	89.3	94.0	87.7	93.8	86.3	93.4	12.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
88.1	204.3	93.6	86.1	191.4	93.0	84.4	181.0	92.9	54.4	94.3	52.1	93.8	50.5	93.4	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.2	31.2	27.1	43.8	114.8	3.4	1.8	41.3	116.1	39.3	117.2	37.4	118.2	34.9	119.5		
			37.4	90.3	3.4	2.8	35.0	92.3	32.9	93.9	31.0	95.5	28.5	97.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	58.6	61.6	68.7	76.6	88.2	96.6	98.8	98.4	94.4	85.9	72.0	62.9
	DBStd	15.20	4.46	4.10	4.35	4.19	4.63	4.01	3.49	3.35	3.44	4.23	5.33	4.00
	HDD50	4	4	0	0	0	0	0	0	0	0	0	0	0
	HDD65	412	201	105	13	0	0	0	0	0	0	0	7	86
	CDD50	11074	270	325	580	798	1183	1398	1512	1501	1333	1112	661	401
	CDD65	6006	1	10	128	348	718	948	1047	1036	883	647	218	22
	CDH74	88525	0	2	281	2216	10179	15975	17962	17660	14368	8812	1067	3
	CDH80	59331	0	0	50	432	5790	11655	13498	13196	10049	4520	141	0
Wind	WSAvg	11.5	11.4	11.0	11.4	10.5	11.3	14.0	12.9	12.2	11.6	10.9	10.5	10.1
Precipitation	PrecAvg	7.2	1.4	0.9	1.2	0.7	0.2	0.0	0.0	0.0	0.0	0.2	1.3	1.2
	PrecMax	11.7	4.1	3.2	3.8	2.0	0.8	0.1	0.1	0.1	0.0	1.0	7.2	3.5
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	1.0	0.8	0.9	0.6	0.2	0.0	0.0	0.0	0.0	0.2	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.9	73.5	85.3	90.5	103.6	112.7	112.5	110.8	107.8	100.6	85.9	73.6
		MCWB	56.7	56.4	59.4	63.7	68.9	69.2	71.2	72.5	71.4	70.6	63.8	59.5
	2%	DB	66.6	70.8	79.8	86.8	99.8	108.8	109.2	108.5	103.9	96.3	82.9	71.0
		MCWB	57.7	56.3	60.3	64.6	69.6	70.4	72.1	72.9	72.1	69.1	69.4	61.0
	5%	DB	65.4	68.7	76.7	84.3	97.3	105.7	107.2	106.4	101.9	93.7	81.6	69.7
		MCWB	58.0	56.4	60.6	65.4	69.5	70.6	72.7	73.5	72.6	70.0	69.1	61.9
	10%	DB	64.3	66.8	74.3	82.3	95.1	103.1	105.2	104.4	100.0	91.8	79.9	68.5
		MCWB	57.6	57.4	60.9	65.5	69.7	70.6	72.5	74.0	73.0	71.8	65.8	60.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	65.2	70.3	74.2	80.9	85.0	90.6	93.3	90.4	84.2	78.3	68.2
		MCDB	65.3	65.8	75.3	81.1	88.5	92.0	92.8	94.1	93.9	89.6	81.8	70.1
	2%	WB	62.0	63.4	67.5	72.5	79.4	82.9	88.3	90.2	87.2	82.4	74.9	66.2
		MCDB	64.6	65.4	72.3	79.0	87.1	92.4	93.9	94.6	93.3	88.6	81.4	69.3
	5%	WB	60.7	61.7	66.5	71.2	78.0	80.4	86.2	88.3	85.8	80.9	71.6	64.5
		MCDB	63.5	64.9	71.1	78.4	86.1	92.2	92.6	94.1	93.0	87.7	78.4	68.2
	10%	WB	59.4	60.5	65.5	70.2	76.6	78.1	84.8	87.2	83.9	79.0	68.9	62.4
		MCDB	62.8	64.4	70.3	77.7	85.7	94.1	92.4	94.0	92.4	86.3	76.3	67.1
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	8.0	7.8	9.5	12.3	12.1	10.3	9.7	8.0	7.9	7.6
		MCDBR	7.5	9.6	12.1	11.5	12.9	15.9	15.0	14.0	12.7	11.5	9.4	8.1
		MCWBR	6.1	5.9	6.6	8.2	8.8	8.7	9.3	9.4	9.6	9.6	8.5	7.0
	5% WB	MCDBR	5.4	5.5	6.2	7.2	7.8	10.7	9.8	7.5	6.2	5.9	6.6	5.1
		MCWBR	7.2	6.1	6.8	8.4	10.2	13.4	12.9	13.4	12.8	12.2	10.8	7.6
Clear-Sky Solar Irradiance	taub	0.511	0.557	0.612	0.699	0.699	0.653	0.678	0.628	0.572	0.576	0.548	0.512	
	taud	1.844	1.752	1.635	1.531	1.535	1.585	1.572	1.662	1.759	1.756	1.801	1.852	
	Ebn at Noon	219	222	221	208	209	218	212	221	228	214	206	211	
	Edh at Noon	56	66	79	91	91	86	87	78	69	65	58	53	
All-Sky Solar Radiation	RadAvg	1078	1348	1704	1857	2143	2387	2288	2175	1936	1538	1125	994	
	RadStd	77	88	74	99	97	80	61	50	66	61	88	90	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.59	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+169	+193

Nomenclature: See separate page